Mohammad Vazir Panah

Embedded Systems Engineer · Embedded Linux, RISC-V, FPGA, C/C++

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Summary

Embedded systems engineer with an M.Sc. in Computer Systems Architecture and three years of R&D on RISC-V and heterogeneous embedded platforms in two multi-partner projects (EuroHPC LIGATE, PRIN LibreRT). I work across the hardware and software boundary: embedded C/C++ and Embedded Linux on RISC-V boards, board bring-up and cross-compilation toolchains, debugging down to the runtime and compiler level, and FPGA design in VHDL and Verilog. Used to reproducible workflows (Git, Docker, CI, CMake) and to international teams.

Technical Skills

Programming	C, C++ (modern C++, templates), Python, Assembly
Embedded Software	Embedded Linux, FreeRTOS, RISC-V boards (AllWinner D1, BananaPi-F3), board bring-up, SoC architecture
Interfaces & Debug	UART, SPI, I2C, JTAG, GDB, low-level runtime debugging
Toolchains & Build	GCC, LLVM/Clang, RISC-V cross-compilation toolchains, CMake
FPGA & Digital Design	VHDL, Verilog, Intel FPGA (DE0-Nano, DE10-Nano-SoC), Quartus, ModelSim, Avalon-MM, high-level synthesis (basic)
Hardware Test	Design for test: full-scan, ATPG, BIST, fault simulation
Parallel & Heterogeneous	OpenMP, OpenCL, SYCL, CUDA
Simulation & Modeling	SystemC, Gem5, QEMU, OVPsim, MATLAB/Simulink
Tools	Git, Docker, CI, Linux
Certification	TinyML, Harvard University (edX)

Professional Experience

Research Fellow, HPC Group (ISISLab) *2023 - 2026*
University of Salerno, Department of Computer Science, Salerno, Italy

- Set up RISC-V cross-compilation toolchains, firmware and Embedded Linux software stacks on physical boards in the EuroHPC LIGATE project, shared across several partner institutions
- Brought up and evaluated RISC-V boards (AllWinner D1, BananaPi-F3) and embedded SoCs for benchmarking and real-time workloads
- Developed and deployed portable real-time applications in C/C++ on heterogeneous embedded CPU/GPU platforms (SYCL, OpenMP, OpenCL) in the LibreRT project
- Found and fixed kernel dispatch failures on Imagination PowerVR embedded GPUs by tracing the issue into the LLVM-based AdaptiveCPP runtime; resolved related OpenCL build problems
- Co-developed an open-source RISC-V benchmark suite (151 loops and 7 real applications) to evaluate GCC and LLVM autovectorization on RISC-V vector hardware; tuning alone gave up to 3× speedup (IEEE PDP 2025)
- Built reproducible build and test pipelines (Git, Docker, CI, CMake) for RISC-V and SoC boards

Teaching Assistant *2020 - 2021*
Shahid Beheshti University, Tehran, Iran

- System-on-Chip, Fundamentals of Embedded & Real-Time Systems, Design & Analysis of Real-Time Embedded Systems, Advanced Computer Architecture
- Ran student FPGA and embedded lab projects from environment setup to lab sessions, feedback and grading

Mandatory Military Service *2016 - 2019*
Iran

C++ Instructor and Hardware Intern *2013 - 2015*
Sepehr Educational Institutions and Abooreyhan Company, Kashmar, Iran

- Taught C++ programming courses; diagnosed and repaired PC hardware, assembled and configured systems

Selected Projects

Real-Time Tiny YOLO Inference and FIR Filter IP Core on FPGA *2019*
Intel DE10-Nano-SoC and DE0-Nano, GitHub

- Implemented a fixed-point real-time Tiny YOLO v1 object-detection pipeline on FPGA under tight resource limits

- Designed a FIR filter IP core in Quartus with an Avalon-MM interface

Custom CPU with Full-Scan DFT, BIST and LSSD Self-Test *2020*
Verilog, GitHub

- Designed a CPU in Verilog and applied scan insertion, fault collapsing and ATPG, reaching 100% fault coverage after scan insertion
- Built LFSR/MISR-based BIST and LSSD boundary-scan self-test for the same design

SADF Model of Computation in ForSyDe-SystemC (M.Sc. Thesis) *2022*
C++ and SystemC, GitHub

- Implemented the Scenario-Aware Dataflow model in C++ templates; the core process constructors were merged into the official ForSyDe-SystemC repository
- Connected the models to ROS and Gazebo to drive a simulated mobile robot in real time

Parallel Sorting and Matrix Multiplication Hardware *2020*
Verilog and CUDA, GitHub

- Designed mesh- and hypercube-connected sorting networks and a shared-memory matrix multiplication architecture in Verilog

Real-Time Audio Controller *2020*
UPPAAL and Lingua Franca, GitHub

- Verified timing properties with timed automata and re-implemented the controller as a reactive program, fixing clock drift between distributed timers

Education

M.Sc. Computer Engineering, Computer Systems Architecture *2019 - 2022*
Shahid Beheshti University, Tehran, Iran

- GPA 18.43/20 · Thesis: *A Framework for Modeling Self-Aware Cyber-Physical Systems with SystemC*

B.Sc. Computer Engineering, Hardware Engineering *2012 - 2016*
Birjand University, Birjand, Iran

- GPA 16/20 · Top 15% of Computer Engineering graduates

Publications

- B. Cosenza et al. (incl. **M. Vazir Panah**). *LibreRT: Portable Heterogeneous Real-Time Programming for the Embedded Computing Continuum.* ACM Computing Frontiers 2026.
- L. Carpentieri, **M. Vazir Panah**, B. Cosenza. *A Performance Analysis of Autovectorization on RVV RISC-V Boards.* IEEE PDP 2025.
- S.-H. Attarzadeh-Niaki, **M. Vazir Panah**. *System-Level Modeling of Dynamic Applications with Scenario-Aware Dataflow Graphs.* JICSE 2023.
- **M. Vazir Panah**, S.-H. Attarzadeh-Niaki, A. Salimi-Badr. *ROS-Based Co-Simulation for Formal Cyber-Physical Robotic System Design.* IEEE CSICC 2022.

Awards

- National R&D Research Grant, Mobile Telecommunication Company of Iran (2020), awarded in competition with Master's and PhD students

Languages

Persian (native) · English (C1) · Italian (B1 listening, A2 speaking and writing)

References available on request.